



Ministry of Health

Arthur Davison Children's Hospital Pathology Laboratory Zambia

Standard Operation Procedure Technical			ID No: ADCH_TP_PCR_005	
			Version: 1	
			Supersedes: 1	
			Effective Date: 04/Nov/19	
			Next Review Date: 04/Nov/21	
TITLE: HIV-GENOTYPING PROCEDURE				
PREPARED BY	NAME	TITLE	SIGNATURE	DATE
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ADCH_PL 	ARTHUR DAVISON CHILDRENS HOSPITAL PATHOLOGY LABORATORY		
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1.0. PURPOSE

HIV-1 genotyping workflow is intended for the detection of HIV-1 genomic mutations in the protease and reverse transcriptase regions of HIV-1 RNA from plasma and dried blood spots. Mutations in the protease and reverse transcriptase regions can confer resistance to specific types of antiretroviral (ARV) drugs.

2.0. SCOPE

To be applied on all samples received in the lab for HIV 1 HIV Drug resistance testing

3.0. RESPONSIBILITY

It is the responsibility of section head and trained, qualified and competent Medical Laboratory Technologist and Scientist.

4.0. PRINCIPLE

This chain-termination (Dye-terminator) method requires a single-stranded DNA template, a DNA primer, a DNA polymerase, normal deoxynucleoside triphosphates (dNTPs), and modified di-deoxynucleotide triphosphates (ddNTPs), the latter of which terminate DNA strand elongation. These chain-terminating nucleotides lack a 3'-OH group required for the formation of a phosphodiester bond between two nucleotides, causing DNA polymerase to cease extension of DNA when a modified ddNTP is incorporated. The ddNTPs are fluorescently labelled for detection in automated sequencing machines.

Sequencing starts by having a short primer binding next to the region of interest. In the presence of the 4 nucleotides, the polymerase extends the primer by adding on the complementary nucleotide from the template DNA strand. To find the exact composition of the DNA sequence, a dideoxynucleotide is added at the end of this particular DNA fragment. The polymerase enzyme can no longer add normal nucleotides onto this DNA chain. The extension has stopped and the chain terminating nucleotide is identified by a specific fluorescent dye, 4 specific colors to be exact. Sanger sequencing results in the formation of

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extension products of various lengths terminated with dideoxynucleotides at the 3' end. The extension products are then separated by Capillary Electrophoresis or CE. The molecules are injected by an electrical current into a long glass capillary filled with a gel polymer (Capillary array). During CE, an electrical field is applied so that the negatively charged DNA fragments move toward the positive electrode. The speed at which a DNA fragment migrates through the medium is inversely proportional to its molecular weight. This process separates the extension products by size at a resolution of one base. A laser excites the dye labelled DNA fragments as they pass through a tiny window at the end of the capillary. The excited dye emits a light at a characteristic wavelength that is detected by a light sensor. Software then interpret the detected signal and translate it into a base call. When the sequencing reaction is performed in the presence of all four terminated nucleotides, a pool of DNA fragments that are measured and separated base by base are obtained.

5.0. SAMPLE REQUIREMENTS

Minimum of 10µl ml of extracted/purified RNA. (Refer to ADCH-XXX-009 on RNA extraction procedure)

6.0. EQUIPMENT

Genetic Analyzer
Thermal cycler
Centrifuge
Vortex mixer
Gel Imager
Gel Electrophoresis equipment
96 well plate mixer

7.0. MATERIALS

POP-7 Polymer 3500 series

ANODE BFFR Container 3500 series

CATHODE BFR Container 3500 series

CONDITIONING REAGNT 3500 series

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HI-DI Formamide

BulkPack 96-well RXN Plates

8-String Optical Tube

Capillary Array 24 X 50CM

HIV-1 GT: Sequencing Module

EXOSAP-IT

E-GEL EX GELS- 2%

0.5 ML PCR tubes

HIV-1 GT: Amplification Module

Nuclease Free Water

Sealing Adhesive Clear Seal

0.2ML Flat Cap Tubes

BIGDYE Xterminator Kit

E-GEL Sample Loading Buffer

Molecular grade ethanol 4L, Absolute (200 proof) Dnase, Rnase, Protease FREE

Pipette tips 0.1-5ml tip sterile

Pipette tips 0.1-10ul Tip sterile

Pipette tips 0.1-20ul Filter Tip sterile

8.0. STORAGE AND STABILITY

Sample type	Temperature	Period
EDTA Whole blood	25 to 30°C (room temperature)	2 hours

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Plasma	-20 to -80°C	Up to 6 weeks
Purified RNA	-15 to -70°C	Up to 1 year

9.0. SAFETY

1	Follow standard laboratory practices while working with biohazardous material.
2	Be sure to wear appropriate protective equipment, including but not limited to, safety glasses, laboratory coats and disposable gloves.
3	If any biohazardous material is spilled, wipe it up immediately and apply disinfectant.
4	If a sample or waste solution comes into contact with your skin, immediately apply disinfectant, and wash it off with soap and water.
5	Do not stare into the laser transmitters beam as your eyesight maybe be severely damaged.

10.0. CALIBRATION

Refer to Applied Biosystems 3500/3500XL Genetic Analyzer User Guide for calibration procedure.

11.0. PROCEDURE STEPS

Perform RT PCR

Set up the RT-PCR reactions

- Prepare sufficient RT-PCR reaction mix for the required number of reactions plus 1 additional reaction for overage, in a chilled tube.

Component	Volume (number of reactions)		
	1	12	24
RT-PCR Master Mix (Normal)	39 µL	468 µL	936 µL
SuperScript™ III One-Step RT-PCR with Platinum™ Taq High Fidelity Enzyme	1 µL	12 µL	24 µL

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Total volume	40 µL	480 µL	960 µL
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- b. Mix the components thoroughly, and then centrifuge briefly to collect contents at the bottom of the tube.
- c. Add 10 µL of RNA or controls to a labeled plate or strip tubes (reaction plate).
Note: Input RNA quantity and quality affect sequencing results.
- d. Denature the RNA and controls in a thermal cycler at 65°C for 10 minutes, then *immediately* place the reaction plate on ice for ≥3 minutes.
- e. Add 40 µL of the RT-PCR reaction mix to each well of the reaction plate.
- f. Vortex gently, then centrifuge briefly to collect contents at the bottom of the wells and immediately proceed to “Run the RTPCR reactions”.

Run the RT-PCR reactions

- a. Load the reaction plate in the thermal cycler.
- b. Set the RT-PCR thermal cycling conditions according to the following table.

Step	Temperature	Time	Cycles
Reverse transcription	50°C	45 minutes	1
Enzyme inactivation	94°C	2 minutes	1
Denature	94°C	15 seconds	40
Anneal	50°C	20 seconds	
Extend	72°C	2 minutes	
Final extension	72°C	10 minutes	1
Hold	4°C	Maximum of 18 hours	

- a. Set the appropriate reaction volume, then start the run.

Perform Nested PCR

Appropriately label the nested PCR plates or strip tubes and place on ice to chill.

- a. Prepare sufficient nested PCR reaction mix for the required number of reactions plus 1 additional reaction for overage, in a chilled tube.

Component	Volume (number of reactions)		
	1	12	24

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Nested PCR Master Mix	47.5 µL	570 µL	1.14 mL
AmpliTaq Gold™ LD DNA Polymerase	0.5 µL	6 µL	12 µL
Total volume	48 µL	576 µL	1.15 L

- b. Mix the components thoroughly, then centrifuge briefly to collect contents at the bottom of the tube and to eliminate air bubbles.
- c. Add 48 µL of the nested PCR reaction mix to each well of a chilled and labeled plate or strip tubes.
- d. Vortex gently, then centrifuge briefly to collect contents at the bottom of the wells.
- e. Add 2 µL of RT-PCR products (including controls) to the labeled plate or strip tubes.
- f. Immediately proceed to “Run the nested PCR reactions”.

Run the nested PCR reactions

- a. Load the reaction plate in the thermal cycler.
- b. Set the nested PCR thermal cycling conditions according to the following table.

Step	Temperature	Time	Cycles
Initial denaturation	94°C	4 minutes	1
Denature	94°C	15 seconds	40
Anneal	55°C	20 seconds	
Extend	72°C	2 minutes	
Final extension	72°C	10 minutes	1
Hold	4°C	Maximum of 18 hours	

- c. Set the appropriate reaction volume, then start the run.

Determine nested PCR product quality

Confirmation of PCR products is recommended to conserve reagents and time.

- a. Visualize the nested PCR products on gel electrophoresis

Note: Use a DNA ladder that includes bands that are near 1 kb in size.

1. Determine if the nested PCR products meet the following criteria:

Sample	Criteria
Positive control	The major product is 1.1 kb

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Negative control	No amplification; no visible DNA bands
Test samples	The major product is 1.1 kb
	No DNA smear

Note: If either of the controls does not meet the criteria, repeat the amplification process.

2. Proceed according to the test sample results:

If the sample displays...	Do this...
No amplification	One of the following (in order) 1. Repeat the RT-PCR and nested PCR 2. Perform a rescue RT-PCR
Low band intensity	One of the following (in order)
Passes the criteria in step 1 above	Proceed to perform sequencing

Perform sequencing

Except as noted, perform cycle sequencing in the Post-amplification laboratory area.

Treat the nested PCR products with ExoSAP-IT PCR Product Cleanup Reagent

- Transfer 10 µL of nested PCR products to a new plate.
- Place the plate and the tube of ExoSAP-IT PCR Product Cleanup Reagent on ice.
- Add 4 µL of ExoSAP-IT PCR Product Cleanup Reagent to each well containing 10 µL of nested PCR products.
- Label the plate "+ExoSAP-IT", then seal the plate with MicroAmp Clear Adhesive Film.
- Vortex the plate for 2 to 3 seconds, then centrifuge at $1,000 \times g$ for 5 to 10 seconds.
- Place the plate into the thermal cycler, then run with the following settings.

Step	Temperature	Time
Digest	37°C	15 minutes
Heat deactivation	80°C	15 seconds
Hold	4°C	

Store the plate on ice for immediate use or at –20°C for longer term storage.

Set up cycle sequencing reactions

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- Completely thaw the six sequencing mixes provided with the HIV-1 Genotyping Kit: Cycle Sequencing Module (F1, F2, F3, R1, R2, R3) on ice. Vortex briefly, then centrifuge for 2–3 seconds to collect contents at the bottom of the tubes.
- Completely thaw the pGEM sequencing control on ice. Vortex briefly, then centrifuge for 2–3 seconds to collect contents at the bottom of the tubes.
- Add 18 µL of each of the six sequencing mixes to the appropriate wells of a chilled 96-well plate.
- Add 2 µL of purified nested PCR products or the positive control to each sequencing mix.
- Add 20 µL of pGEM Sequencing Control to at least one well per run.
- Seal the plate and immediately proceed to “Run the cycle sequencing reactions” on the sequencing conditions below.

Step	Temperature	Time	Cycles
Denature	96°C	10 seconds	25
Anneal	50°C	5 seconds	
Extend	60°C	4 minutes	
Hold	4°C	Maximum of 18 hours	

- Set the appropriate reaction volume, then start the run.
(Optional) Store sequencing products at –20°C for up to 3 days.

Purify sequencing reactions with the BigDye XTerminator Purification Kit

- Remove the XTerminator Solution from 4°C storage and allow it to equilibrate to room temperature.
- Vortex the XTerminator Solution for at least 10 seconds before mixing with the SAM solution.
- Prepare the SAM/BigDye XTerminator bead working solution:

Component	Volume per 20-µL reaction	Volume per 96-well plate
SAM™ solution	90 µL	9.9 ml
XTerminator™ solution	20 µL	2.2 ml
Total volume	110 µL	12.1 ml

- Remove the MicroAmp Clear Adhesive Film from the sequencing plate.

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- e. Dispense 110 µL/well of the SAM/BigDye XTerminator bead working solution to each sample. To mix thoroughly, pipette the solution up and down 3–4 times before each transfer.
- f. Seal the plate using MicroAmp Clear Adhesive Film.
- g. Vortex the 96-well plate for 30 minutes at 1,800 rpm.
- h. Centrifuge the plate at 1,000 × *g* in a swinging bucket centrifuge for 2 minutes at room temperature.
- i. Proceed immediately to capillary electrophoresis.

(Optional) Sequencing reactions purified with the BigDye XTerminator Purification Kit can be stored at 4°C overnight if necessary.

Run capillary electrophoresis

- a. Perform maintenance activities if required. (Refer to instrument User Guide, ADCH_EXT_059)
- b. Remove the adhesive film, if present, then replace with a 96-well plate septa.
- c. Load the plate into the genetic analyzer.
- d. Select the 50-cm capillary length, the number of capillaries, and the POP-7™ polymer type.
- e. Select or create an appropriate run module according to instrument user guide.
- f. Start the run.

Determine sequence quality

- a. Open the data collection software and review the .ab1 files for quality control parameters. Refer to instrument user guide for standard procedures for sequence detection and analysis.
- b. If traces are of good quality, proceed to “Generate consensus sequences” with either Sequencher or RECall software.
- c. Review consensus sequence calls and ensure variants were not missed. Edit the basecalls as needed.

Data Management

- a. Once the sequences are analyzed on the Genetic Analyzer, create a folder (separate from the “Sequence Data” folder linked to the Genetic Analyzer) with folders listed by date Year (20xx)-Month(xx)-Day(xx).
- b. Within each folder containing the date of the sample run, create folders with each patient ID (Full national ID number-10 digit Accession Number). ~As a reminder, the primary data

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files (.ab1) from the genetic analyzer should be labeled as follows “Full accession number (3 letter prefix plus 7 digits)-primer number

- c. Copy and transport the .AB1 raw data files from the “Data” Folder into the respective patient folder. Do not move the original files
- d. Convert the original files into one analyzable by Sequencer. Open Sequencing Analyzer 5.3.1: (Refer to Sequencer Software Overview, (ADCH_EXT_058)
- e. Analyze converted/compatible .AB1 files on the Sequencer Software (Details in Refer to Sequencer Software Overview, EXT PCR 0002)
- f. To save sequence analysis file of the Sample, the consensus sequences must now be entered into the Stanford Database (<http://hivdb.stanford.edu/index.html>) to determine sequence quality and subtype.
- g. Print the result to PDF and save by PID in the matching Project file.
- h. Once they are reviewed result can be reported.
- i. After reviewing and editing the entire sequence, the following three files need to be saved in the appropriate folders (in Completed Reports):
- j. Converted **.AB1** files. Depending on the number of sequences run, there can be multiple files with different primer numbers. (named “3 letter prefix plus 7 digits accession-primer.number.AB1”)
- k. Project file **.SPF**. Entire analysis project generated on Sequencer. (Named “3 letter prefix plus 7 digits accession.SPF”). Please note that within the .SPF project file, the actual contig that gets generated must also be named with the same nomenclature because this is how the fasta.TXT file gets its header information within the file.
- l. Fasta File stored in **.TXT**. This is the contig that was analyzed on sequencer and exported out of the .SPF project file (named “3 letter prefix plus 7 digits accession.TXT”)
- m. PDF file stored as **.PDF**. This is the pdf report from the fasta.txt file that was uploaded to Stanford drug resistance database.

Batch analysis

- a. To perform batch analysis of patient data stored in fasta.txt files for research purposes.
- b. Collect all fasta.txt files into one folder. Then, compile all fasta files with the following commands:
- c. For Macintosh Operating System, open the Terminal prompt then enter: “**cat** * > NAME.txt” [where * is the name of the directory for the stored files and NAME.txt is the name of the compiled fasta text file]
- d. Windows Operating System, open the command line then go to the directory where your

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files are contained. (To change directories in command line, first type "cd" then click and drag a file from the folder that contains all text files, delete the name of the file and just remain with the folder name)

- e. Then once you are in that directory, type "copy *.txt NAME.txt" [where * is the name of the directory and NAME.txt is the name of the compiled fasta text file]
- f. Upload the batch fasta file onto the Stanford database using the same Sequence Anaysis program that was used for single patient analysis (<http://sierra2.stanford.edu/sierra/servlet/JSierra?action=sequenceInput>), but instead of the "html output" option in the "Output Analysis" box, choose "Spreadsheet output" to get your mutations listed as a binary output.
- g. Wait for the analysis to be completed (depending on how many samples you analyze, this can take several minutes)
- h. Copy and paste the characters onto excel sheet (note, Google's Chrome browser makes the does not work and you should use Mozilla Firefox instead). Paste as special as "text"
- i. Determine if there are any quality assurance issues (ie, ambiguous bases, Frame shifts resulting in stop codons, (B,D,H,V,N) triplicate nucleotide ambiguity, missing amino acids, etc.)
- j. This excel sheet can be integrated with other clinical information (medication history) as well as other relevant biomarkers retrieved from LIMS (ie, CD4 counts, viral load at the time of collection, etc.)
- k. Excel output:

Columns B-AD: Quality control measures for Protease, Reverse Transcriptase, and Integrase regions. Note that Integrase columns are blank because Integrase is not a part of our assay.

Columns AE-AW: Protease resistance profile.

Columns AX-BV: Reverse Transcriptase resistance profile

Columns BW-CC: Integrase resistance profile

Columns CD-LI: Protease resistance binary coding

Columns LJ-YC: Reverse transcriptase resistance binary coding

Columns YD-AGX: Integrase resistance binary coding

12.0. REFERENCE RANGES

N/A

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13.0. INTERPRETATION OF THE RESULTS

For...	Use the...
Individual mutation analysis	Online ReCall software. Go to http://pssm.cfenet.ubc.ca . or Stanford HIV Resistance database/HIVdb Program to generate resistance data and mutation reports.
Population-based surveillance sequences	Stanford Calibrated Population Resistance (CPR) tool to generate resistance mutation reports for transmitted resistance. Go to http://sierra2.stanford.edu/sierra/servlet/JSierra?action=sequenceInput or http://cpr.stanford.edu/cpr .

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14.0. LIMITATIONS OF THE PROCEDURE AND SOURCES OF ERROR

The HIV-1 population within an individual consists of innumerable variants and minor variants that often go undetected

15.0. REPORTING OF RESULTS

"**Susceptible**" indicates no evidence of reduced ARV susceptibility compared with a wild-type virus.

"**Potential low-level resistance**" indicates that the sequence may contain mutations indicating previous ARV exposure or may contain mutation that are associated with drug resistance only when they occur with additional mutations.

"**Low-level resistance**" indicates that there that the virus encoded by the submitted sequence may have reduced in vitro ARV susceptibility or that patients harboring viruses with the submitted mutations may have a suboptimal virological response to treatment with the ARV.

"**Intermediate resistance**" indicates a high likelihood that a drug's activity will be reduced but that the drug will likely retain significant remaining antiviral activity.

"**High-level resistance**" indicates that the predicted level of resistance is similar to those observed in viruses with the highest levels of in vitro drug resistance or that clinical data exist demonstrating that patients infected with viruses having such mutations usually have little or no virological response to treatment with the ARV.

16.0. SUPPORTING DOCUMENTS

- a. Applied Biosystems 3500/3500xL Genetic Analyzer User Guide
- b. Sequencer 5 Series User Guide for Mac and Windows

17.0. REFERENCES

- a. Stanford University HIV Drug resistance <https://hivdb.stanford.edu/hivdb/by-sequences/>
- b. *ThermoFisher Scientific Applied Biosystems HIV 1 Genotyping Workflow*

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18.0. RECORDS

RECORD TITLE	RECORD ID	CUSTODIAN/LOCATION
Sample request /report form		Section head-Virology
Applied Biosystems Maintenance form		Section head-

19.0. AMENDMENT RECORD

NAME	DATE	SUMMARY OF CHANGES

10. CONTROL COPY DISTRIBUTION LOG

Copies To:	Number of copies issued:	Received By (Initials):	Received Date:	Obsolete Version Returned By:
QA MASTER FILE				
VIROLOGY				
MICROBIOLOGY/PARASITOLOGY				
HAEMATOLOGY				
BTS/SEROLOGY				
CHEMISTRY				

[illegible]